TOPIC 01 · FUNDAMENTALS OF GOOGLE SHEET
LAB 2 THEME · DATA QUALITY CHECK

Lab 2 — Check the Quality of the Candidate Data with Gemini

Maps to	LO1: use Gemini to profile HR data for duplicates, inconsistent categories, missing or invalid values and wrong data types, highlight the problems and confirm each count with a formula
Duration	25 minutes
Tools	Google Sheets · Gemini side panel (questions, conditional-formatting actions, formulas) · OrchidBay_Q3_Applications_QualityCheck.xlsx
Mock data	data/OrchidBay_Q3_Applications_QualityCheck.xlsx
Course	Gemini for Google Sheet (C817) · v2.0
Prompts	https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-2

Goal

Find, highlight and count every data-quality problem in the Q3 export before anyone analyses it - and prove each count with a formula.

What You Will Do

Before the recruitment manager sees any numbers, you check the raw Q3 export. You will ask Gemini to profile the data, find the repeated Candidate_IDs and the Source spelling variants, and apply conditional formatting actions that highlight duplicates, bad emails, text salaries and impossible values. A Checks tab of formulas confirms every count, and Gemini drafts a short data-quality report. You do not fix anything yet - that is Lab 3.

What You Will Build

OrchidBay_Q3_Applications_QualityCheck with duplicates and problem cells highlighted, a Checks tab of formulas confirming each issue count, an inserted data-quality report and a filled Prompt_Log

Prerequisites

- Lab 1 completed, or at least the Gemini side panel opened once.
- This lab folder downloaded, including data/OrchidBay_Q3_Applications_QualityCheck.xlsx.

Data note

Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Steps

Step 1. Download OrchidBay_Q3_Applications_QualityCheck.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

PROMPT / CHECK — copy and adapt

```
Required state:
File : OrchidBay_Q3_Applications_QualityCheck (Google Sheets copy)
Tabs : About_This_Data | Applications | Jobs | Checks | Prompt_Log
Rule : find and flag only - do not fix anything in this lab
```

Step 2. Ask Gemini to profile the data for the common problems. Read its list carefully: it is a starting point, not the final answer - the formulas in Step 6 decide.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Check the Applications tab for data-quality problems: duplicate Candidate_IDs, the same Source
spelt different ways, missing or invalid emails, Expected_Salary values stored as text, and
impossible values.
Don't change anything. List each problem with how many rows it affects.
```

Step 3. Ask about duplicates, then apply Gemini's conditional-formatting action. An Action preview card appears - read it and click Apply. When it finishes, Settings and Undo appear: Settings opens the rule, Undo reverses it. There are 243 rows but only 240 unique IDs: CAN-26531, CAN-26601, CAN-26608 each appear twice, so 6 cells turn orange (press Ctrl + Down arrow to reach the copies near the end).

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Which Candidate_IDs appear more than once? Are those rows exact copies of each other?

ACTION:
Highlight duplicate values in the Candidate_ID column in orange.
```

Step 4. Ask Gemini to list Source exactly as spelt. Besides the seven real sources you should see 'Linked In' (3), 'linkedin' (2) and 'JobStreet ' with a trailing space (3). Spelling variants split one source into several and corrupt every count by Source.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
List every distinct value in the Source column exactly as it is spelt, including spaces and
capital letters, with a count of each.
```

Step 5. Send these one change at a time. After each Action preview card, click Apply, look at the sheet, and use Undo if it is not what you wanted. Expect 6 email cells (4 blank, 2 without @), 5 text salaries and 1 negative Years_Experience (CAN-26607).

PROMPT / CHECK — copy and adapt

```
ACTION:
1) Highlight cells in the Email column that are blank or do not contain @ in light red.
2) Highlight Expected_Salary cells that contain text instead of a number in yellow.
3) Highlight Years_Experience values below 0 in red.
```

Step 6. Now prove each count. The Checks tab already has the labels in column A and the expected values in column C. First ask Gemini for the unique-ID formula and insert it into Checks!B3. Then paste the other check formulas into column B beside their labels. Every B must match its C; if Gemini's Step 2 numbers disagree, the formulas win.

PROMPT / CHECK — copy and adapt

```
FORMULA:
Create a formula that counts the unique Candidate_IDs in Applications column A, rows 2 to 244.

CHECK:
Total rows =COUNTA(Applications!A2:A244)
Unique IDs =COUNTUNIQUE(Applications!A2:A244)
Blank emails =COUNTBLANK(Applications!C2:C244)
Emails without @ =COUNTIFS(Applications!C2:C244,"<>",Applications!C2:C244,"<>*@*")
Text salaries =SUMPRODUCT(--ISTEXT(Applications!K2:K244))
Negative experience =COUNTIF(Applications!H2:H244,"<0")
```

Step 7. Ask Gemini for a short report for the recruitment manager, then click Insert - Gemini adds the table to your spreadsheet (usually on a new tab); rename that tab DQ_Report. Or use Copy and paste it into an email draft.

PROMPT / CHECK — copy and adapt

```
INSERT:
Write a short data-quality report for the recruitment manager: each issue, how many rows it
affects, why it matters for screening, and the fix you recommend. Use a table.
```

Step 8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

PROMPT / CHECK — copy and adapt

```
LOG:
List the prompts I sent you in this conversation. For each one, give a one-line summary of
what you did, so I can copy it into the Prompt_Log tab.
```

Sample Prompts

Use these prompts in the Gemini side panel in Google Sheets (or adapt them). Name the tab and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online: https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-2

#	Purpose	Prompt
1	Profile	Which columns have blank cells, and how many blanks does each have?
2	Validate	Are there any Applied_Date values outside July to September 2026?
3	Validate	Do any rows have a Position that doesn't match the Position for their Job_ID on the Jobs tab?
4	Highlight	Highlight Full_Name cells that are all in capital letters in light purple.
5	Explain	Why is a salary stored as text a problem when I calculate an average?

Test It

Check	Expected result
Duplicates	243 rows, 240 unique IDs; CAN-26531, CAN-26601, CAN-26608 each appear twice; 6 orange cells.
Source variants	'Linked In' 3, 'linkedin' 2, 'JobStreet ' (trailing space) 3.
Emails	4 blank + 2 without @ (CAN-26697, CAN-26606) = 6 light-red cells.
Salaries and experience	5 text salaries in yellow; 1 negative Years_Experience (CAN-26607) in red.
Checks tab	Six formulas in column B match the expected values already in column C (243, 240, 4, 2, 5, 1).

Checkpoint

Lab 3 opens a fresh copy of the same raw export and fixes every issue you found here.

Troubleshooting

If this happens	Fix
There is no Apply button - Gemini only explains how.	Rephrase as a direct instruction ('Highlight ... in orange'), and make sure the Applications tab is selected. If actions are not available on your plan, use Format > Conditional formatting with Gemini's steps.
Gemini highlights the whole row instead of the cell.	Click Undo and say 'only the cells in the Candidate_ID column'.
COUNTUNIQUE returns 243.	Check the range: it must be A2:A244 on Applications, not a range that includes the header twice.
Gemini's counts differ from the formulas.	Ask Gemini how it counted. Trust the formula and refine the prompt.

Challenge

Add a formula that counts the rows whose Position does not match the Position of their Job_ID on the Jobs tab (hint: ask Gemini for a SUMPRODUCT with VLOOKUP). The answer should be 0.

Reflection

Which issue would most distort a 'hires by source' report if nobody had caught it?